

In-frame deletion of lasB in LB001 via pEXG2 allelic exchange

Primer design • PCR / Assembly protocol • 2026-05-06

1 Konstrukt-Überblick

Zielgen	lasB in LB001 (1497 nt, 498 aa)
Strategie	In-frame Deletion, Scar 18 aa (16 N + 2 C)
Vektor	pEXG2 (5084 bp), HindIII-linearisiert
Methode	In-Fusion 3-Fragment (Vektor + UP + DN)
Counterselektion	sacB / GmR (pEXG2)
Finales Plasmid	pEXG2-ΔlasB_LB001 • 6155 bp

2 Primer

Primer	Sequenz 5'→ 3' (Tail fett, Body normal)	nt	Tm _{body}	GC
P1_lasB_UP_Fwd	CATAAATGTAAAGCAGGTGTTCCAGCTGGTGCAG	34	60.9 °C	63 %
P2_lasB_UP_Rev	CCGAGCTTACAACGCACCCATGATCGCAACGAACAAC	37	60.8 °C	50 %
P3_lasB_DN_Fwd	GTTGCGATCATGGGTGCGTTGTAAGCTCGGTGGTC	35	61.0 °C	60 %
P4_lasB_DN_Rev	CGACCTGCAGAAGCTGCCAGGTACTCGCCTTGC	33	60.7 °C	67 %

3 PCR-Bedingungen (B7 High Fidelity)

Tm _{body} Mittel	60.9 °C (Spread 0.21 °C)
Annealing Ta	61 °C (= Tm _{body} ; Gradient ± 5 °C empfohlen falls 1. Versuch fehlschlägt)
Elongation	60 s/kb • UP 593 bp → 36 s • DN 538 bp → 32 s

4 Konstrukt-Zusammenfassung

UP-Amplicon 593 bp • DN-Amplicon 538 bp • Insert 1101 bp • Final 6155 bp

Scar-Sequenz (16 N + 2 C aa + Stop):

ATG AAG AAG GTT TCT ACG CTT GAC CTG TTG TTC GTT GCG ATC ATG GGT | GCG TTG TAA
M K K V S T L D L L F V A I M G | A L *

Off-Target Scan: PASS • HindIII-Sites in finalem Plasmid: 0 (erwartet 0)